



Inspiring Excellence

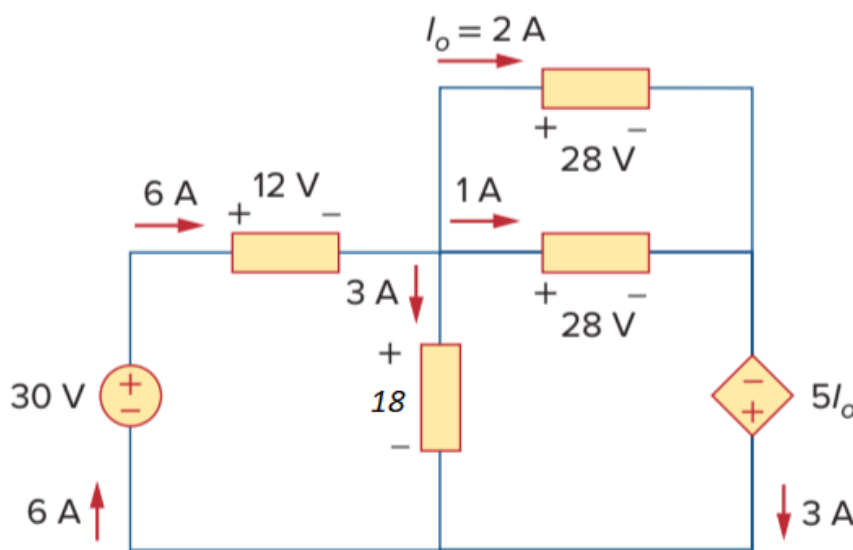
**Department of CSE**  
**CSE250: Circuits and Electronics**  
**Assignment-1**

**Full Marks: 35 + 5 bonus points**

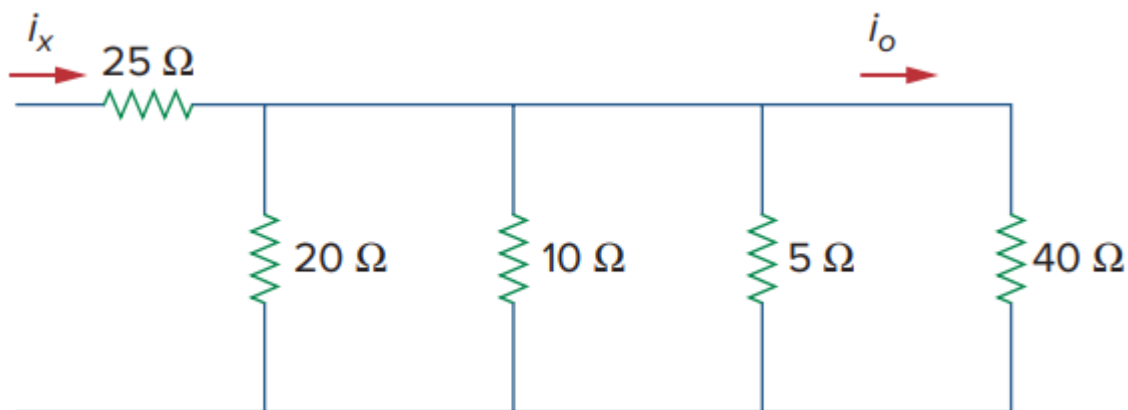
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Each of the questions falls inside Course Outcome 1 and Program Outcome 1.

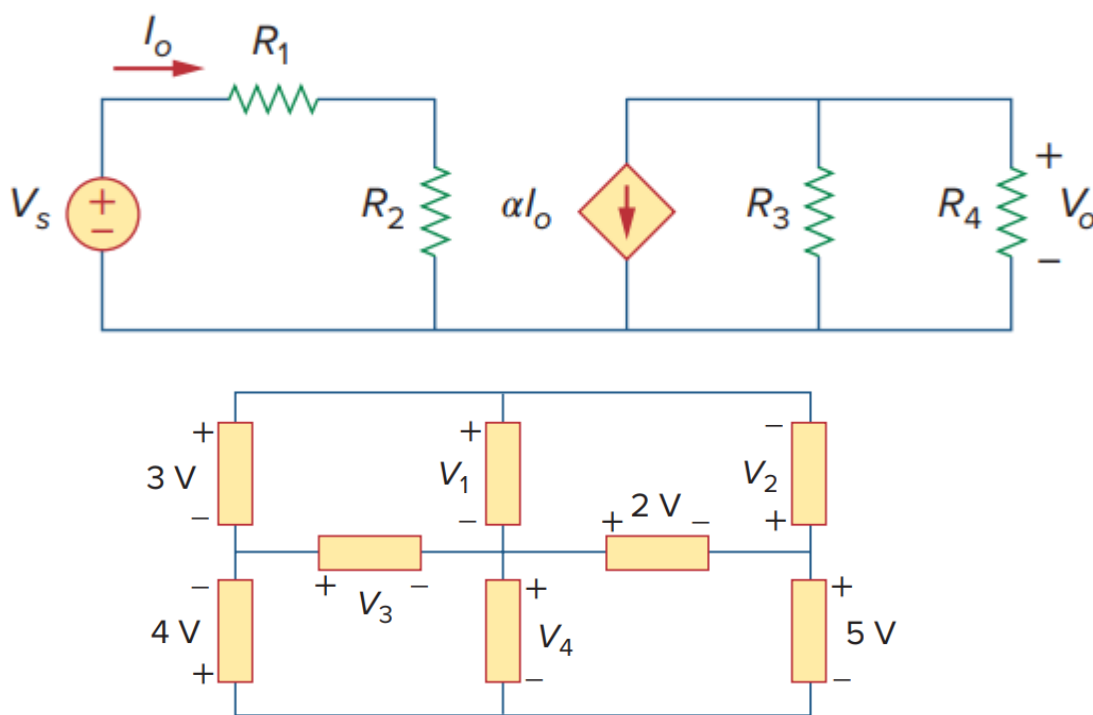
1. 10 J work is done on a  $-2\text{C}$  charge in moving it from point A to point B, where  $V_B = 20\text{ V}$ . **Solve** for the potential of point A. (1)
2. The total charge entering a terminal is given by  $q = (2 - Ae^{-At/2})\text{ mC}$ . **Identify** the current at  $t = 2\text{ s}$ . Here, A = last 2 digits of your student ID. (2)
3. **Identify** the power supplied/absorbed by each of the elements shown in the circuit below. (6)



4. Here,  $i_o = X\text{ A}$ , where X = last two digits of your roll. **Identify**  $i_x$ . (5)

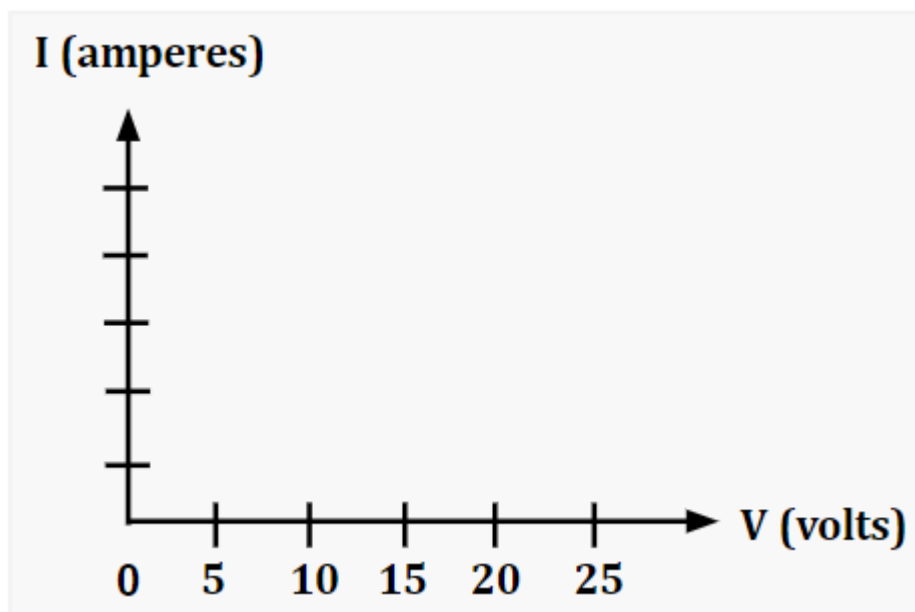


5. In the following figure,  $R_1 = R_3 = (1 + \text{last digit of your student ID}) \text{ k}\Omega$ ,  
 $R_2 = 2R_4 = (2 + \text{second digit of your student ID}) \text{ k}\Omega$ ,  
 $V_s = (3 + \text{last digit of your student ID}) \text{ V}$ ,  
 $\alpha = \text{sum of first 2 digits of your student ID}$ .  
**Solve** for  $I_o$ ,  $V_o$ , and voltage across  $R_2$ . (6)

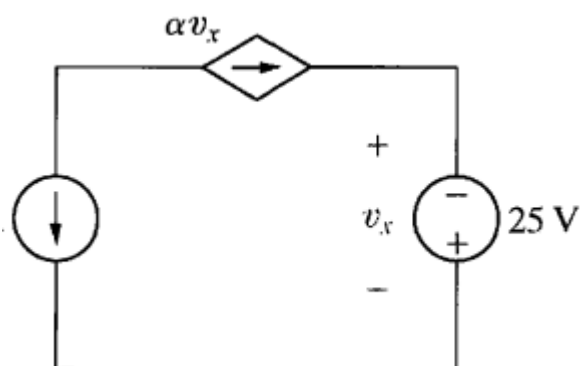


6. Given the circuit below, **use** KVL to find the branch voltages  $V_1$  to  $V_4$ . (4)

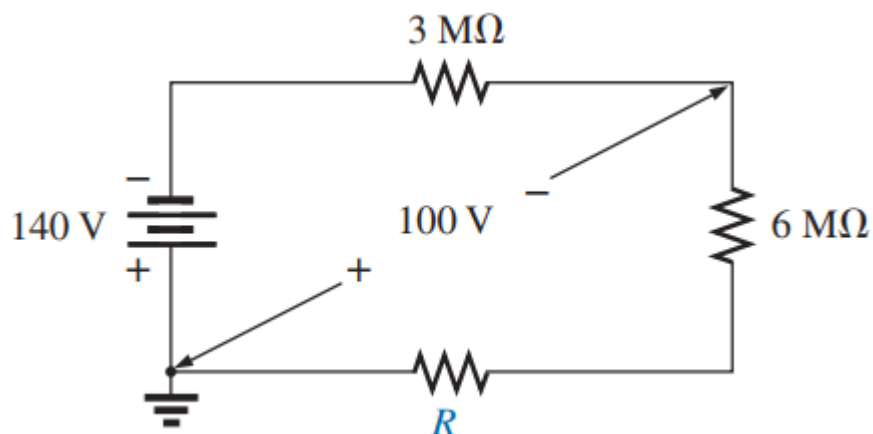
7. **Sketch** the I-V characteristic of a  $B \text{ k}\Omega$  resistor using the following template.  $B$  = sum of last 2 digits of your student ID. (3)



8. Here, the value of the current source is  $Y \text{ V}$ , where  $Y$  = sum of first 3 digits of your student ID. Find the value of  $\alpha$  required for the interconnection to be validly **operated**. (3)



9. **Implementing** the voltage divider rule, find the unknown resistance for the configuration below. (5)



10. **(BONUS)**

Find  $v_1$  and  $v_g$  in the circuit shown when  $v_o$  equals Z V, where Z = last two digits of your student ID. **Execute** Ohm's Law. (Hint: Start at the right end of the circuit and work back toward  $v_g$ ). (5)

